



Off by One

Quick facilitation notes & answer key

Goal

Students find and fix off-by-one loop bugs (C3.2) in both block (count) and turtle (a square that does not close) contexts, and describe how the one-step change fixes the outcome.

Curriculum (Ontario Math 2020, Strand C3)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential, concurrent, and repeating events.

C3.2 — read and alter existing code, including code that involves sequential, concurrent, and repeating events, and describe how changes to the code affect the outcomes.

Materials

A block-coding app and/or Python Turtle projected; the printed worksheet + a pencil per student.

Run it (about 30 min)

1. Run a turtle square with `range(3)` — the open shape makes off-by-one visible.
2. Add one to range — it closes. Name the bug: 'off by one.'
3. Students do Exercises 1–4 (Ex 4 is the stretch), solo or in pairs.
4. Share answers; land the big idea: check the count against the goal every time.

Answer key (checked by the run-gate)

Ex 1 — one too many; repeat 5.

Ex 2 — too few; repeat 6.

Ex 3 — `range(4)`.

Ex 4 (challenge) — only 3 sides (an open shape); change `range(3)` to `range(4)` to close the square.

Watch for & differentiate

Common slip: reading `range(4)` as 5 (counting the start) — it is exactly 4.

Support: count the sides on the drawing and compare to the range number.

Extend: ask the off-by-one fix for a triangle drawn with `range(2)` (range 3).

Success looks like

The student identifies too-many vs too-few and corrects the count (Ex 1–3), and explains how adding one side closes the square (Ex 4).